

# AP Statistics MCQ Practice Exam 03

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## AP Statistics MCQ Practice Exam 3

### PART A, QUESTIONS

#### APSTAT-MCQ3-001

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

The lengths, in millimeters, of five microplastic fibers are 3, 5, 7, 9, and 11. What is the sample standard deviation of these lengths?

- A) 2.00 mm
- B) 2.83 mm
- C) 10.00 mm
- D) 3.16 mm

#### APSTAT-MCQ3-002

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Formulate Questions

Prompt:

Which research question can be addressed using a one-proportion z-confidence interval?

- A) What is the mean weekly screen time of all students at a university?
- B) What proportion of all registered boats in a state have a current safety inspection?
- C) How much do the proportions of urban and rural residents who compost differ?
- D) What is the mean change in grip strength among patients after physical therapy?

#### APSTAT-MCQ3-003

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

In a large population, 40% of members have a digital library card. For a random sample of 100 members, what is the approximate probability that more than 50% have a digital card? Use  $P(Z > 2.04) = 0.0207$ .

- A) 0.0207
- B) 0.0400
- C) 0.4793
- D) 0.9793

APSTAT-MCQ3-004

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

Independent random samples of 18 ceramic glazes and 24 glass glazes are used for a two-sample t-procedure comparing population mean curing times. If software is unavailable and the conservative degrees-of-freedom rule is used, what value of df should be used?

- A) 40
- B) 23
- C) 17
- D) 16

APSTAT-MCQ3-005

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A city survey asks, "Do you agree that responsible residents should conserve water whenever possible?" Which revision would best reduce wording-related response bias while still measuring the intended behavior?

- A) "Responsible residents conserve water; do you?"
- B) "Why do you choose to conserve water?"
- C) "During the past seven days, on how many days did you take an action specifically to reduce household water use?"
- D) "Would you admit that you sometimes waste water?"

APSTAT-MCQ3-006

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Interpret Results

Prompt:

A scatterplot of daily traffic volume and average vehicle speed has a clear curved, inverted-U pattern, but the correlation is  $r = 0.04$ . Which interpretation is most appropriate?

- A) The correlation is near zero because correlation measures linear association and can miss a strong curved relationship.
- B) The variables have no relationship because  $r$  is close to zero.
- C) The curved pattern proves that traffic volume causes changes in speed.
- D) The correlation must be incorrect because every visible pattern has  $|r|$  close to 1.

APSTAT-MCQ3-007

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A two-sided test of a population proportion produces a p-value of 0.073. Which decision is correct?

- A) Reject  $H_0$  at  $\alpha = 0.05$  and at  $\alpha = 0.10$ .
- B) Fail to reject  $H_0$  at  $\alpha = 0.05$  and at  $\alpha = 0.10$ .
- C) Reject  $H_0$  at  $\alpha = 0.05$  but fail to reject  $H_0$  at  $\alpha = 0.10$ .
- D) Fail to reject  $H_0$  at  $\alpha = 0.05$  but reject  $H_0$  at  $\alpha = 0.10$ .

APSTAT-MCQ3-008

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

Questions APSTAT-MCQ3-008 through APSTAT-MCQ3-010 use the following stimulus. Across 30 wetlands,  $x$  is mean spring nighttime temperature in degrees Celsius and  $y$  is the mean number of moth species observed per trap-night. The summary statistics are  $\bar{x} = 12^\circ\text{C}$ ,  $s_x = 2^\circ\text{C}$ ,  $\bar{y} = 18$  species,  $s_y = 5$  species, and  $r = 0.60$ .

Which point must lie on the least-squares regression line for predicting  $y$  from  $x$ ?

- A) (2, 5)
- B) (12, 18)
- C) (18, 12)
- D) (30, 0.60)

APSTAT-MCQ3-009

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

Questions APSTAT-MCQ3-008 through APSTAT-MCQ3-010 use the following stimulus. Across 30 wetlands,  $x$  is mean spring nighttime temperature in degrees Celsius and  $y$  is the mean number of moth species observed per trap-night. The summary statistics are  $\bar{x} = 12^\circ\text{C}$ ,  $s_x = 2^\circ\text{C}$ ,  $\bar{y} = 18$  species,  $s_y = 5$  species, and  $r = 0.60$ .

For a wetland with  $x = 14^\circ\text{C}$ , what number of moth species does the least-squares line predict?

- A) 18 species
- B) 19 species
- C) 21 species
- D) 23 species

APSTAT-MCQ3-010

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Interpret Results

Prompt:

Questions APSTAT-MCQ3-008 through APSTAT-MCQ3-010 use the following stimulus. Across 30 wetlands,  $x$  is mean spring nighttime temperature in degrees Celsius and  $y$  is the mean number of moth species observed per trap-night. The summary statistics are  $\bar{x} = 12^\circ\text{C}$ ,  $s_x = 2^\circ\text{C}$ ,  $\bar{y} = 18$  species,  $s_y = 5$  species, and  $r = 0.60$ .

A wetland with  $x = 14^\circ\text{C}$  is 1 standard deviation above the mean of  $x$ , but its predicted  $y$ -value is only 0.60 standard deviation above the mean of  $y$ . What feature of least-squares prediction does this illustrate?

- A) Predicted  $y$ -values always have a larger standard deviation than observed  $y$ -values.
- B) A positive correlation forces every residual to be positive.
- C) The regression line predicts  $y$  to be the same number of raw units above its mean as  $x$  is above its mean.

D) When  $|r| < 1$ , predicted standardized responses are closer to their mean than the corresponding standardized x-values.

#### APSTAT-MCQ3-011

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Formulate Questions

Prompt:

Which setting can reasonably be modeled with a binomial random variable?

- A) Count the number of correct answers among 15 independent true-or-false questions when the probability of a correct answer is 0.50 on each question.
- B) Count correct answers on 15 quiz questions when the questions have different numbers of answer choices.
- C) Record the total value of 10 independent rolls of a fair die.
- D) Count red cards in 20 draws without replacement from a standard deck.

#### APSTAT-MCQ3-012

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

The reported mean mass of 30 shipping crates is 42.5 kilograms. One mass was entered as 72 kilograms but should have been 27 kilograms. What is the corrected mean mass?

- A) 39.5 kg
- B) 41.0 kg
- C) 42.0 kg
- D) 44.0 kg

#### APSTAT-MCQ3-013

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Interpret Results

Prompt:

For the same sample mean, sample standard error, and 95% confidence level, how does a t-confidence interval with  $df = 9$  compare with a z-confidence interval?

- A) The t-interval is narrower because estimating  $\sigma$  reduces uncertainty.
- B) The t-interval is wider because the t critical value accounts for uncertainty from estimating  $\sigma$  with a small sample.
- C) The intervals have equal width because both use a 95% confidence level.
- D) The t-interval is shifted upward because the t-distribution has mean greater than zero.

APSTAT-MCQ3-014

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Collect Data

Prompt:

A researcher asks employees about an embarrassing workplace mistake. To use randomized response, each employee privately flips a fair coin and answers “yes” if it lands heads; if it lands tails, the employee answers the sensitive question truthfully. What is the main statistical benefit of this design?

- A) It makes the sample a census of all employees.
- B) It removes all sampling variability from the estimated proportion.
- C) It identifies which individual employees made the mistake.
- D) It can reduce dishonest responses while still allowing the population proportion to be estimated after accounting for the forced “yes” answers.

APSTAT-MCQ3-015

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Each of five smart locks activates successfully and independently with probability 0.92. What is the probability that at least one of the five locks fails to activate?

- A) 0.0000033
- B) 0.0800
- C) 0.3409
- D) 0.6591

APSTAT-MCQ3-016

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A botanist places all plants receiving blue light in one growth chamber and all plants receiving red light in a second chamber. Many plants are placed in each chamber. Why does this design not permit a clear causal comparison of light color?

- A) Light color is confounded with any chamber-specific difference, because there is only one chamber for each treatment.
- B) The experiment has too much replication because several plants receive each color.
- C) A causal conclusion requires the plants to choose their own light condition.
- D) Measuring plant growth quantitatively prevents random assignment.

APSTAT-MCQ3-017

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

Historical data indicate that the population standard deviation of fill volume for a beverage machine is  $\sigma = 12$  milliliters. A planner uses  $z^* = 1.96$  and wants a 95% confidence interval for the population mean with margin of error at most 3 milliliters. What minimum sample size is required?

- A) 16
- B) 31
- C) 61
- D) 62

APSTAT-MCQ3-018

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

A random sample of eligible voters produces a 95% confidence interval of (0.52, 0.60) for the population proportion who support a ballot measure. Which conclusion is justified?

- A) Exactly 56% of all eligible voters support the measure.
- B) Ninety-five percent of eligible voters have support probabilities between 0.52 and 0.60.
- C) The interval provides evidence that more than half of eligible voters support the measure.
- D) There is a 95% probability that the fixed population proportion changes from 0.52 to 0.60.

APSTAT-MCQ3-019

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

For a dataset relating robotic-arm speed to placement error, the least-squares slope is negative and  $r^2 = 0.49$ . What is the correlation?

- A)  $-0.49$
- B)  $-0.70$
- C)  $0.49$
- D)  $0.70$

APSTAT-MCQ3-020

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A continuous random measurement is modeled by a uniform density curve from 10 to 30 units. What is the probability that a randomly selected measurement is between 14 and 23 units?

- A) 0.45
- B) 0.55
- C) 0.65
- D) 0.90

APSTAT-MCQ3-021

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

A population of invoice amounts has mean  $\mu = \$120$  and standard deviation  $\sigma = \$18$ . For random samples of  $n = 81$  invoices, the sampling distribution of  $\bar{x}$  is approximately normal. What is  $P(\$117 < \bar{x} < \$123)$ ? Use  $\Phi(1.50) = 0.9332$  and  $\Phi(-1.50) = 0.0668$ .

- A) 0.0668
- B) 0.4332
- C) 0.8664

D) 0.9332

#### APSTAT-MCQ3-022

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

In independent random samples, 84 of 120 customers shown package A and 72 of 120 customers shown package B recognized the brand. Which is the approximate 90% confidence interval for  $p_A - p_B$ ? Use  $z^* = 1.645$ .

- A) (0.039, 0.161)
- B) (0.000, 0.200)
- C) (-0.001, 0.201)
- D) (-0.021, 0.221)

#### APSTAT-MCQ3-023

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Collect Data

Prompt:

A medical researcher recruits 30 pairs of twins to compare two physical-therapy routines. Within each twin pair, one twin is randomly assigned routine A and the other routine B. The response is improvement in shoulder range of motion. Which analysis respects the design?

- A) A one-proportion procedure using the proportion of twins who improve
- B) A paired t-procedure applied to the within-pair differences in improvement
- C) A one-sample t-procedure using only the routine A improvements
- D) A two-sample t-procedure that ignores the twin pairings

#### APSTAT-MCQ3-024

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Interpret Results

Prompt:

The 65th percentile of customer wait time at a service center is 12 minutes. Which statement correctly interprets this percentile?

- A) Sixty-five percent of customers wait exactly 12 minutes.
- B) The mean wait is 12 minutes for 65% of customers.
- C) Thirty-five percent of customers wait no more than 12 minutes.
- D) About 65% of customers wait 12 minutes or less, and about 35% wait longer than 12 minutes.

APSTAT-MCQ3-025

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Questions APSTAT-MCQ3-025 through APSTAT-MCQ3-027 use the following stimulus. Let  $X$  be the number of electric-vehicle charging ports in use at a randomly selected time. The cumulative probability distribution is:

|               |      |      |      |      |      |
|---------------|------|------|------|------|------|
| $x$           | 0    | 1    | 2    | 3    | 4    |
| $P(X \leq x)$ | 0.10 | 0.35 | 0.70 | 0.90 | 1.00 |

What is  $P(X = 2)$ ?

- A) 0.35
- B) 0.45
- C) 0.65
- D) 0.70

APSTAT-MCQ3-026

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Questions APSTAT-MCQ3-025 through APSTAT-MCQ3-027 use the following stimulus. Let  $X$  be the number of electric-vehicle charging ports in use at a randomly selected time. The cumulative probability distribution is:

|               |      |      |      |      |      |
|---------------|------|------|------|------|------|
| $x$           | 0    | 1    | 2    | 3    | 4    |
| $P(X \leq x)$ | 0.10 | 0.35 | 0.70 | 0.90 | 1.00 |

What is  $P(X \geq 3)$ ?

- A) 0.10

- B) 0.20
- C) 0.70
- D) 0.30

### APSTAT-MCQ3-027

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Interpret Results

Prompt:

Questions APSTAT-MCQ3-025 through APSTAT-MCQ3-027 use the following stimulus. Let  $X$  be the number of electric-vehicle charging ports in use at a randomly selected time. The cumulative probability distribution is:

|               |      |      |      |      |      |
|---------------|------|------|------|------|------|
| $x$           | 0    | 1    | 2    | 3    | 4    |
| $P(X \leq x)$ | 0.10 | 0.35 | 0.70 | 0.90 | 1.00 |

Which statement correctly identifies and interprets the median of  $X$ ?

- A) The median is 1 because  $P(X \leq 1)$  is closest to 0.50.
- B) The median is 2 because at least half the probability is at or below 2 and at least half is at or above 2.
- C) The median is 3 because  $P(X \leq 3) = 0.90$ .
- D) The median is 0.70 because cumulative probability first exceeds 0.50 at  $x = 2$ .

### APSTAT-MCQ3-028

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Formulate Questions

Prompt:

A subscription service wants to estimate the difference between the cancellation proportions for customers under age 30 and customers age 30 or older. Independent random samples will be taken from the two age groups. Which procedure is appropriate?

- A) A one-proportion z-test
- B) A paired t-interval
- C) A two-proportion z-confidence interval
- D) A two-sample t-confidence interval

### APSTAT-MCQ3-029

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A state agency wants to estimate household broadband availability across a large rural region. It randomly selects 12 census blocks, then takes a simple random sample of 15 households within each selected block. Which description best fits the design?

- A) A multistage sample: clusters are selected first, followed by random sampling within selected clusters
- B) A stratified sample: every census block is represented by 15 households
- C) A simple random sample of all households in the region
- D) A voluntary-response sample within randomly chosen census blocks

APSTAT-MCQ3-030

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

For a two-sided one-sample t-test with  $df = 9$ , the observed statistic is  $t = 2.262$ . A t-table shows  $P(T \geq 2.262) = 0.025$  for  $df = 9$ . What is the p-value?

- A) 0.050
- B) 0.025
- C) 0.950
- D) 0.975

APSTAT-MCQ3-031

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Formulate Questions

Prompt:

A retailer wants to use weekly online-ad impressions to predict weekly demand for a product. Which assignment of variables is appropriate for a regression analysis?

- A) Demand is  $x$  because it is more variable, and ad impressions are  $y$ .
- B) Both variables are response variables because both are quantitative.

- C) Ad impressions and demand can be assigned randomly to  $x$  and  $y$  without changing the prediction goal.
- D) Ad impressions are the explanatory variable  $x$ , and demand is the response variable  $y$ .

#### APSTAT-MCQ3-032

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A manufacturer tests  $H_0: p = 0.02$ , where  $p$  is the population proportion of smoke detectors that fail a sensitivity check, and uses  $\alpha = 0.01$ . Which statement describes  $\alpha$ ?

- A) If  $H_0$  is false, the test has a 1% chance of detecting the false value.
- B) If  $H_0$  is true, the long-run probability that the test rejects  $H_0$  is 0.01.
- C) After the data are observed, the probability that  $H_0$  is true is 0.01.
- D) One percent of all tested smoke detectors must fail the sensitivity check.

#### APSTAT-MCQ3-033

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A histogram of daily grocery-delivery orders over one year has two clear peaks, one near 85 orders and another near 145 orders, with relatively few days near 115 orders. The peaks correspond mainly to weekdays and weekends. Which summary is most informative?

- A) Report only the overall mean because it lies between the two peaks.
- B) Describe the distribution as approximately normal because there are two symmetric peaks.
- C) Report the two modes and compare weekday and weekend distributions separately.
- D) Remove all weekend days as outliers before describing the distribution.

#### APSTAT-MCQ3-034

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Collect Data

Prompt:

A printing engineer wants to estimate the probability that a product label is readable by both scanner A and scanner B. Which data-collection plan best preserves the joint outcomes needed for this probability?

- A) Randomly select labels, test every selected label with both scanners in randomized order, and record the paired outcomes for each label.
- B) Use one label sample for scanner A and an unrelated sample for scanner B, then multiply the two sample success proportions.
- C) Test only labels that have already passed a visual inspection and record whether either scanner reads them.
- D) Ask production staff to estimate the two separate readability probabilities from experience.

#### APSTAT-MCQ3-035

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Interpret Results

Prompt:

A one-sample  $t$ -interval for mean component lifetime is based on  $n = 8$  observations. A dotplot shows seven values between 40 and 55 hours and one value at 130 hours. Which assessment is most appropriate?

- A) The interval is reliable because  $t$ -procedures are unaffected by all outliers.
- B) The interval is reliable because  $n$  is less than 30.
- C) The extreme value should automatically be deleted before any investigation.
- D) With such a small sample, the severe outlier threatens the  $t$ -procedure; its cause should be investigated and more data may be needed.

#### APSTAT-MCQ3-036

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Collect Data

Prompt:

An online retailer randomly assigns website visitors to checkout page A or page B and compares purchase proportions. Some visitors return several times during the study and can be assigned again. What is the main concern?

- A) Random assignment makes a comparison of proportions impossible.
- B) Outcomes from repeat visits by the same person may not be independent, so the visit should not automatically be treated as an independent experimental unit.
- C) A quantitative response is required for a randomized experiment.
- D) The study must use equal purchase proportions under both pages.

APSTAT-MCQ3-037

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Formulate Questions

Prompt:

Which is a statistical question that anticipates variability and can be answered by collecting quantitative data?

- A) What is the serial number of this repaired appliance?
- B) Did one particular elevator stop at floor 6 today?
- C) How does the distribution of repair times differ among three appliance models?
- D) Is the posted elevator capacity exactly 1,200 kilograms?

APSTAT-MCQ3-038

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Let  $X$  have a binomial distribution with  $n = 60$  and  $p = 0.05$ . Which description of the distribution of  $X$  is most appropriate?

- A) Approximately normal because  $n \geq 30$
- B) Approximately uniform because all counts from 0 to 60 are possible
- C) Symmetric because the mean is  $np = 3$
- D) Right-skewed because  $np = 3$  is small while  $n(1 - p) = 57$

APSTAT-MCQ3-039

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

A sunscreen manufacturer tests  $H_0: p = 0.90$  against  $H_a: p > 0.90$ , where  $p$  is the population proportion of users protected for the labeled duration. The test gives  $p$ -value 0.18. At  $\alpha = 0.05$ , which conclusion is appropriate?

- A) Fail to reject  $H_0$ ; the data do not provide convincing evidence that the protection proportion exceeds 0.90.

- B) Accept  $H_0$  and conclude that exactly 90% of all users are protected for the labeled duration.
- C) Reject  $H_0$  because 0.18 is greater than 0.05.
- D) Conclude that 18% of users are not protected for the labeled duration.

#### APSTAT-MCQ3-040

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Interpret Results

Prompt:

A licensing agency takes a random sample from all currently licensed drivers in a state and estimates their mean hazard-perception score. Which population is directly supported by the sample design for generalization?

- A) Every adult resident of the state
- B) All currently licensed drivers in the state
- C) Only the licensed drivers who completed the test
- D) All future applicants for a driver's license

#### APSTAT-MCQ3-041

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

A 92% confidence interval for a population mean is  $48.6 \pm 3.4$ . If the same data are used to construct a 95% confidence interval, which change should occur?

- A) The center decreases and the margin of error stays 3.4.
- B) The center increases and the margin of error decreases.
- C) The center remains 48.6 and the margin of error increases.
- D) The center remains 48.6 and the margin of error decreases.

#### APSTAT-MCQ3-042

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

Output voltage from a model of solar inverter is approximately normal with mean 230 volts and standard deviation 4 volts. By the empirical rule, approximately what proportion of outputs are between 222 and 238 volts?

- A) 0.95
- B) 0.68
- C) 0.997
- D) 1.00

## PART B, ANSWER KEY AND EXPLANATIONS

APSTAT-MCQ3-001, Correct: D, Correct Answer: 3.16 mm, Explanation: The sample mean is 7 mm. The squared deviations sum to  $16 + 4 + 0 + 4 + 16 = 40$ , so  $s = \sqrt{40/(5 - 1)} = \sqrt{10} \approx 3.16$  mm.

APSTAT-MCQ3-002, Correct: B, Correct Answer: What proportion of all registered boats in a state have a current safety inspection?, Explanation: This question concerns one population proportion. Choice C concerns a difference between two proportions, while choices A and D concern means.

APSTAT-MCQ3-003, Correct: A, Correct Answer: 0.0207, Explanation: The sampling distribution has mean 0.40 and standard deviation  $\sqrt{0.40(0.60)/100} \approx 0.0490$ . Thus  $z = (0.50 - 0.40)/0.0490 \approx 2.04$ , so  $P(\hat{p} > 0.50) \approx 0.0207$ .

APSTAT-MCQ3-004, Correct: C, Correct Answer: 17, Explanation: The conservative rule uses the smaller of  $n_1 - 1$  and  $n_2 - 1$ . Thus  $df = \min(18 - 1, 24 - 1) = \min(17, 23) = 17$ .

APSTAT-MCQ3-005, Correct: C, Correct Answer: "During the past seven days, on how many days did you take an action specifically to reduce household water use?", Explanation: The revision asks about a defined behavior during a specific period without labeling one response as responsible or irresponsible. The other phrasings are leading or assume conservation occurred.

APSTAT-MCQ3-006, Correct: A, Correct Answer: The correlation is near zero because correlation measures linear association and can miss a strong curved relationship., Explanation: Correlation summarizes linear direction and strength. Positive and negative linear tendencies in an inverted-U pattern can offset, producing  $r$  near zero even when a clear non-linear association exists.

APSTAT-MCQ3-007, Correct: D, Correct Answer: Fail to reject  $H_0$  at  $\alpha = 0.05$  but reject  $H_0$  at  $\alpha = 0.10$ ., Explanation: Reject  $H_0$  when  $p\text{-value} \leq \alpha$ . Since  $0.073 > 0.05$  but  $0.073 < 0.10$ , the decisions differ at the two significance levels.

APSTAT-MCQ3-008, Correct: B, Correct Answer: (12, 18), Explanation: Every least-squares regression line passes through  $(\bar{x}, \bar{y})$ , which here is (12, 18).

APSTAT-MCQ3-009, Correct: C, Correct Answer: 21 species, Explanation: A temperature of  $14^\circ\text{C}$  is  $(14 - 12)/2 = 1$  standard deviation above  $\bar{x}$ . The predicted response is  $r(1) = 0.60$  standard deviation above  $\bar{y}$ , so  $\hat{y} = 18 + 0.60(5) = 21$  species.

APSTAT-MCQ3-010, Correct: D, Correct Answer: When  $|r| < 1$ , predicted standardized responses are closer to their mean than the corresponding standardized x-values., Explanation: In standardized units,

the regression prediction is  $z_{\hat{y}} = r z_x$ . With  $r = 0.60$  and  $z_x = 1$ ,  $z_{\hat{y}} = 0.60$ , illustrating regression toward the mean.

APSTAT-MCQ3-011, Correct: A, Correct Answer: Count the number of correct answers among 15 independent true-or-false questions when the probability of a correct answer is 0.50 on each question., Explanation: This setting has a fixed number of independent trials, two outcomes per trial, and constant success probability. The other settings have changing success probabilities, count a total rather than successes, or use dependent draws without replacement.

APSTAT-MCQ3-012, Correct: B, Correct Answer: 41.0 kg, Explanation: The reported total is  $30(42.5) = 1,275$  kg. Correcting 72 to 27 reduces the total by 45 kg, so the corrected mean is  $(1,275 - 45)/30 = 41.0$  kg.

APSTAT-MCQ3-013, Correct: B, Correct Answer: The t-interval is wider because the t critical value accounts for uncertainty from estimating  $\sigma$  with a small sample., Explanation: With the standard error and confidence level held fixed, the  $df = 9$  t critical value exceeds the corresponding z critical value. The interval therefore has a larger margin of error but the same center.

APSTAT-MCQ3-014, Correct: D, Correct Answer: It can reduce dishonest responses while still allowing the population proportion to be estimated after accounting for the forced “yes” answers., Explanation: The researcher cannot link an individual “yes” to the sensitive behavior because it may have been forced by the coin. In aggregate, the known 0.50 forced-yes probability can be removed when estimating the truthful proportion.

APSTAT-MCQ3-015, Correct: C, Correct Answer: 0.3409, Explanation: The complement of “at least one fails” is “all five activate.” Thus  $P(\text{at least one failure}) = 1 - (0.92)^5 \approx 1 - 0.6591 = 0.3409$ .

APSTAT-MCQ3-016, Correct: A, Correct Answer: Light color is confounded with any chamber-specific difference, because there is only one chamber for each treatment., Explanation: Any difference in growth could be caused by light color or by temperature, humidity, or another chamber feature. Multiple plants within one chamber do not provide independent treatment replication across chambers.

APSTAT-MCQ3-017, Correct: D, Correct Answer: 62, Explanation: Require  $1.96(12/\sqrt{n}) \leq 3$ . Solving gives  $n \geq (1.96 \times 12/3)^2 = 61.47$ , so the sample size must be rounded up to 62.

APSTAT-MCQ3-018, Correct: C, Correct Answer: The interval provides evidence that more than half of eligible voters support the measure., Explanation: Every plausible value in the interval is above 0.50, supporting a population majority. The interval estimates one fixed population proportion, not individual probabilities.

APSTAT-MCQ3-019, Correct: B, Correct Answer:  $-0.70$ , Explanation: Since  $r^2 = 0.49$ ,  $|r| = \sqrt{0.49} = 0.70$ . The negative regression slope determines the negative sign, so  $r = -0.70$ .

APSTAT-MCQ3-020, Correct: A, Correct Answer: 0.45, Explanation: For a uniform density curve, probability is the fraction of total interval length. The desired length is  $23 - 14 = 9$  and the total length is  $30 - 10 = 20$ , so the probability is  $9/20 = 0.45$ .

APSTAT-MCQ3-021, Correct: C, Correct Answer: 0.8664, Explanation: The standard deviation of  $\bar{x}$  is  $18/\sqrt{81} = \$2$ . The bounds have z-scores  $-1.50$  and  $1.50$ , so the probability is  $0.9332 - 0.0668 = 0.8664$ .

APSTAT-MCQ3-022, Correct: C, Correct Answer:  $(-0.001, 0.201)$ , Explanation:  $\hat{p}_A = 0.70$ ,  $\hat{p}_B = 0.60$ , and the difference is 0.10. The unpooled standard error is  $\sqrt{0.70(0.30)/120 + 0.60(0.40)/120} \approx 0.06124$ . Thus  $0.10 \pm 1.645(0.06124) \approx 0.10 \pm 0.1007 = (-0.001, 0.201)$ .

APSTAT-MCQ3-023, Correct: B, Correct Answer: A paired t-procedure applied to the within-pair differences in improvement, Explanation: Twin pairs create matched observations, and random assignment occurs within each pair. Analyzing within-pair differences preserves the blocking and controls pair-to-pair variation.

APSTAT-MCQ3-024, Correct: D, Correct Answer: About 65% of customers wait 12 minutes or less, and about 35% wait longer than 12 minutes., Explanation: The 65th percentile is a value at or below which about 65% of observations fall. The remaining approximately 35% are above that value.

APSTAT-MCQ3-025, Correct: A, Correct Answer: 0.35, Explanation:  $P(X = 2) = P(X \leq 2) - P(X \leq 1) = 0.70 - 0.35 = 0.35$ .

APSTAT-MCQ3-026, Correct: D, Correct Answer: 0.30, Explanation:  $P(X \geq 3) = 1 - P(X \leq 2) = 1 - 0.70 = 0.30$ .

APSTAT-MCQ3-027, Correct: B, Correct Answer: The median is 2 because at least half the probability is at or below 2 and at least half is at or above 2., Explanation:  $P(X \leq 1) = 0.35 < 0.50$ , while  $P(X \leq 2) = 0.70 \geq 0.50$ . Also  $P(X \geq 2) = 1 - 0.35 = 0.65$ , so 2 satisfies the median criterion.

APSTAT-MCQ3-028, Correct: C, Correct Answer: A two-proportion z-confidence interval, Explanation: The target is a difference between two population proportions based on independent samples. A confidence interval, rather than a test, is appropriate because the goal is estimation.

APSTAT-MCQ3-029, Correct: A, Correct Answer: A multistage sample: clusters are selected first, followed by random sampling within selected clusters, Explanation: Sampling occurs in two stages: census blocks are selected, then households are randomly selected within chosen blocks. It is not stratified because not every block is represented.

APSTAT-MCQ3-030, Correct: A, Correct Answer: 0.050, Explanation: The supplied probability 0.025 is one tail beyond  $|t| = 2.262$ . A two-sided p-value includes both tails, so  $p\text{-value} = 2(0.025) = 0.050$ .

APSTAT-MCQ3-031, Correct: D, Correct Answer: Ad impressions are the explanatory variable  $x$ , and demand is the response variable  $y$ ., Explanation: The prediction goal uses ad impressions as the input and demand as the outcome. Swapping the variables changes the least-squares prediction problem.

APSTAT-MCQ3-032, Correct: B, Correct Answer: If  $H_0$  is true, the long-run probability that the test rejects  $H_0$  is 0.01., Explanation: The significance level is the probability of a Type I error under the null. It is not a posterior probability that  $H_0$  is true or a statement about the observed sample percentage.

APSTAT-MCQ3-033, Correct: C, Correct Answer: Report the two modes and compare weekday and weekend distributions separately., Explanation: The overall distribution is bimodal because it combines two meaningful groups. An overall mean near the low-frequency gap can be unrepresentative, so group-specific descriptions are more informative.

APSTAT-MCQ3-034, Correct: A, Correct Answer: Randomly select labels, test every selected label with both scanners in randomized order, and record the paired outcomes for each label., Explanation: The probability of readability by both scanners requires paired joint outcomes on the same labels. Separate samples provide marginal readability rates but do not reveal their dependence or joint probability.

APSTAT-MCQ3-035, Correct: D, Correct Answer: With such a small sample, the severe outlier threatens the t-procedure; its cause should be investigated and more data may be needed., Explanation: One-sample t methods are not robust to a severe outlier when  $n = 8$ . The point should not be deleted automatically, but data quality and the population model should be investigated.

APSTAT-MCQ3-036, Correct: B, Correct Answer: Outcomes from repeat visits by the same person may not be independent, so the visit should not automatically be treated as an independent experimental unit., Explanation: Repeated outcomes from one visitor can be correlated. Random page assignment does not by itself justify treating multiple visits from the same person as independent observations.

APSTAT-MCQ3-037, Correct: C, Correct Answer: How does the distribution of repair times differ among three appliance models?, Explanation: Repair time is quantitative, the wording anticipates variation across observations, and data can be collected to compare distributions. The other choices ask about a fixed fact or one event.

APSTAT-MCQ3-038, Correct: D, Correct Answer: Right-skewed because  $np = 3$  is small while  $n(1 - p) = 57$ , Explanation: A binomial distribution is approximately normal only when both  $np$  and  $n(1 - p)$  are sufficiently large. With only 3 expected successes, most probability is near small counts with a tail toward larger counts.

APSTAT-MCQ3-039, Correct: A, Correct Answer: Fail to reject  $H_0$ ; the data do not provide convincing evidence that the protection proportion exceeds 0.90., Explanation: Since  $0.18 > 0.05$ , the result is not statistically significant. Failing to reject  $H_0$  does not prove  $p = 0.90$ , and the p-value is not a sample failure proportion.

APSTAT-MCQ3-040, Correct: B, Correct Answer: All currently licensed drivers in the state, Explanation: Random sampling supports generalization to the population from which the sample was drawn: current state-licensed drivers. It does not automatically cover unlicensed adults or future applicants.

APSTAT-MCQ3-041, Correct: C, Correct Answer: The center remains 48.6 and the margin of error increases., Explanation: The point estimate remains the same when confidence level changes. Greater confidence requires a larger critical value, which makes the interval wider.

APSTAT-MCQ3-042, Correct: A, Correct Answer: 0.95, Explanation: The interval 222 to 238 volts is  $230 \pm 8$ , or within 2 standard deviations of the mean. For an approximately normal distribution, the empirical rule places about 95% of observations within 2 standard deviations.

#### SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-MCQ3

· APSTAT-MCQ3-001 to 005: Unit 1 five microplastic-fiber lengths compute sample standard deviation from squared deviations and  $n - 1$ ; Unit 3 registered-boat inspections select a research question suited to a one-proportion confidence interval; Unit 2 digital-library-card sampling calculates an upper-tail probability for  $\hat{p}$  from its sampling distribution; Unit 4 ceramic-versus-glass glaze curing samples apply the conservative two-sample t degrees-of-freedom rule; Unit 1 water-conservation survey rewrites a leading question as a neutral behavior-frequency item. Misconceptions include using population rather

than sample variance, confusing one proportion with means or a difference of proportions, using  $p$  rather than the sampling standard deviation, adding sample degrees of freedom, and retaining socially desirable wording.

- APSTAT-MCQ3-006 to 010: Unit 5 curved traffic-volume versus speed scatterplot explains why  $r$  near zero can coexist with a strong non-linear association; Unit 3  $p$ -value 0.073 compares decisions at  $\alpha = 0.05$  and 0.10; Unit 5 three-question wetland stimulus uses  $\bar{x}$ ,  $s_x$ ,  $\bar{y}$ ,  $s_y$ , and  $r$  to identify the point  $(\bar{x}, \bar{y})$ , predict moth-species count for an  $x$ -value one standard deviation above its mean, and interpret standardized regression toward the mean. Misconceptions include treating  $r = 0$  as no relationship, reversing the  $p$ -value decision rule, assuming the regression line passes through the origin, and predicting equal standardized distances when  $|r| < 1$ .

- APSTAT-MCQ3-011 to 015: Unit 2 independent true-or-false answers identify the fixed-trial binary constant-probability binomial setting; Unit 1 shipping-crate masses correct a mean after replacing a miscoded observation through total adjustment; Unit 4 compares  $df = 9$   $t$  and  $z$  interval widths at fixed standard error; Unit 3 employee randomized response uses a private coin flip to reduce dishonesty while preserving aggregate estimation; Unit 2 smart-lock activations calculate at-least-one failure through the all-success complement. Misconceptions include accepting varying-probability or without-replacement counts as binomial, changing only the erroneous value without updating the total, assuming  $t$  and  $z$  critical values coincide, expecting randomized response to identify individuals, and multiplying failure probabilities for an at-least-one event.

- APSTAT-MCQ3-016 to 020: Unit 1 one-growth-chamber-per-light-color plant experiment diagnoses chamber-treatment confounding despite many plant replicates; Unit 4 beverage fill-volume planning uses known  $\sigma$ ,  $z^*$ , a target margin, and upward rounding; Unit 3 ballot-measure one-proportion interval entirely above 0.50 supports a majority conclusion; Unit 5 robotic-arm speed and placement error recover negative  $r$  from negative slope and  $r^2 = 0.49$ ; Unit 1 uniform density from 10 to 30 computes interval probability by relative length. Misconceptions include pseudoreplication, rounding required  $n$  downward, treating a confidence interval as individual probabilities, ignoring slope sign when taking  $\sqrt{r^2}$ , and using normal-curve rules for a uniform density.

- APSTAT-MCQ3-021 to 025: Unit 2 invoice sample means use  $\sigma/\sqrt{n}$  and symmetric  $z$ -bounds to find a central probability; Unit 3 package-recognition samples construct an unpooled 90% interval for  $p_A - p_B$ ; Unit 4 twin pairs with treatment randomized within pairs select a paired  $t$  analysis on improvement differences; Unit 1 service-center wait time interprets a 65th percentile; Unit 2 charging-port cumulative distribution begins by differencing adjacent cumulative probabilities to recover  $P(X = 2)$ . Misconceptions include using  $\sigma$  instead of the standard error, pooling proportions for a confidence interval, ignoring matching, treating a percentile as a mean or exact frequency, and reading a cumulative entry as a point probability.

- APSTAT-MCQ3-026 to 030: Unit 2 charging-port cumulative distribution uses a complement for  $P(X \geq 3)$  and locates the discrete median; Unit 3 subscription cancellations choose a two-proportion  $z$ -interval for estimation; Unit 1 rural broadband uses a multistage census-block-then-household design; Unit 4 two-sided  $t$ -test doubles a supplied one-tail probability. Misconceptions include confusing  $\geq$  with a cumulative  $\leq$  entry, reporting probability rather than the median's  $x$ -value, choosing a test when estimation is requested, calling selected clusters strata, and failing to double a one-tail area.

- APSTAT-MCQ3-031 to 035: Unit 5 online-ad impressions and product demand assign explanatory and response roles from a prediction goal; Unit 3 smoke-detector failure testing interprets  $\alpha = 0.01$  as a conditional long-run Type I error rate; Unit 1 bimodal grocery-delivery orders tied to weekdays and weekends motivate group-specific descriptions; Unit 2 product labels tested by two scanners preserve paired joint outcomes to estimate readability by both devices; Unit 4 component lifetimes with  $n = 8$  and a severe high outlier evaluate  $t$ -procedure robustness. Misconceptions include treating  $x$  and  $y$  as interchangeable, reading  $\alpha$  as  $P(H_0)$ , summarizing a mixture only by its mean, multiplying marginal estimates without dependence information, and assuming  $t$  methods are outlier-proof.

- APSTAT-MCQ3-036 to 040: Unit 3 checkout-page experiment recognizes dependence from repeat visits by the same person; Unit 1 appliance repair times identify a statistical question that anticipates quantitative variability; Unit 2 Binomial(60, 0.05) uses  $np = 3$  and  $n(1 - p) = 57$  to identify right skew rather than a normal approximation; Unit 3 sunscreen-duration right-tailed test with p-value 0.18 gives a fail-to-reject conclusion without accepting  $H_0$ ; Unit 1 random sample of currently licensed drivers limits generalization to its sampling population. Misconceptions include treating visits as independent experimental units, confusing fixed factual questions with statistical questions, invoking  $n \geq 30$  alone for binomial normality, interpreting p-value as an observed failure rate, and generalizing beyond the sampling frame.
- APSTAT-MCQ3-041 to 042: Unit 4 changing a mean interval from 92% to 95% confidence holds the center fixed and increases the margin of error; Unit 1 approximately normal solar-inverter voltage uses the empirical rule for the interval  $\mu \pm 2\sigma$ . Misconceptions include shifting the point estimate when confidence changes, believing higher confidence narrows an interval, and confusing the 68%, 95%, and 99.7% empirical-rule bands.
- Unit counts for APSTAT-MCQ3: Unit 1 = 11; Unit 2 = 9; Unit 3 = 9; Unit 4 = 7; Unit 5 = 6. Skill counts: Formulate Questions = 5; Collect Data = 7; Analyze Data = 21; Interpret Results = 9. Stimulus sets: wetland nighttime-temperature and moth-species standardized regression, APSTAT-MCQ3-008 to 010; electric-vehicle charging-port cumulative distribution, APSTAT-MCQ3-025 to 027.